

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Flash memory microcontroller

Renesas Electronics announces an on-chip flash memory microcontroller (MCU) using a 28nm process technology. To contribute to the realisation of next-generation green cars and autonomous vehicles with higher efficiency and reliability, RH850/E2xSeries MCU incorporates up to six 400MHz CPU cores, which makes it the first on-chip flash memory automotive MCU to achieve a processing performance of 9600MIPS. The MCU series also features a built-in flash memory of up to 16MB as well as enhanced security functions and functional safety.

Renesas Electronics Corp.
www.renesas.com

Optical rotary torque sensors

Sensor Technology launches digital ORT 230/240 series optical rotary torque sensors that are ideal for applications that need low torque and/or high bandwidth, providing precise, dynamic measurement of rotary and static torque of less than 100nm and for bandwidths of up to 50kHz. The ORT 230/240 devices replace Sensor Technology's E200 ORT series, benefiting from all-new electronics that deliver significant gains in resolution, frequency response, reduced sensor current consumption and faster digital data throughput.

Sensor Technology Ltd
www.sensors.co.uk

Chip capacitors

TDK has developed a new series of vertically-stacked MEGACAP type multilayer ceramic chip capacitors that combine high capacitance and low ESR.

The series offers rated voltages from 25V to 1000V, and covers a capacitance range from 20nF to 150µF. The capacitors are available with C0G, X7T, X7S and X7R temperature characteristics. Owing to their high capacitance values, these are suitable for resonant circuits of wireless and plug-in charging systems; for example, for industrial vehicles and robots. These can also be used in smoothing and decoupling applications in industrial equipment.

TDK Corp.
www.global.tdk.com



Microcontrollers

Designers of Internet of Things (IoT) sensors, environmental sensors, smartwatches, medical/preventive health wearables and other size-constrained devices can now increase battery life and functionality using the ultra-low power MAX32660 and MAX32652 MCUs from Maxim Integrated Products. These MCUs are based on Arm Cortex-M4 with FPU processor and provide designers the means to develop advanced applications under restrictive power constraints.

Maxim Integrated
www.maximintegrated.com

Image sensors

ON Semiconductor introduces the first CMOS image sensors to feature its Near Infra-Red + (NIR+) technology, which effectively combines high dynamic range with enhanced low-light performance to enable high-end security and surveillance cameras. AR0522 is a 5.1MP image sensor based on 2.2µm back side illuminated (BSI) pixel technology platform, developed for indus-



trial applications that require high resolution, high-quality video capture in low light conditions.

AR0431 is a 4MP sensor based on 2.0µm BSI pixel technology platform. Offering versatile low power modes and a frame rate of up to 120fps, it is perfect for applications where slow-motion video capture is required.

ON Semiconductor
www.onsemi.com

Digital-power controller

STMicroelectronics simplifies and accelerates development of 90W-300W power supplies and adaptors for computers, LED lighting, medical equipment, telecom and industrial equipment, meeting today's most stringent eco-design standards. The controller combines a double-ended LLC resonant half-bridge controller and a multimode power-factor correction controller, managed by a digital core running best-in-class control algorithms. Integrated digital peripherals leveraging STMicroelectronics' state-machine-event-driven technology and proprietary analogue hardware accelerate control-loop performance for excellent dynamic response.

STMicroelectronics India
www.st.com



Switching regulators

Traco Power announces the new TSR 2 series of step-down switching regulators that are drop-in replacement for LMxx linear regulators. A high efficiency up to 96 per cent allows full load operation up to +67°C ambient temperature (full-load) without any heat-sink or forced cooling.

The regulators provide significant features over linear regulators including better output accuracy ($\pm 2\%$), lower standby current of 1mA and no

